

# VaR, CVaR/TVaR, and Expected Shortfall: A Practitioner's Reference

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COMPANION TO *ERQI* — *Model Risk Testing & Analysis for Cyber Risk Quantification*

Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR/TVaR) are standard financial risk measures used across insurance, banking, and enterprise risk management. They are not specific to cyber risk — they apply identically to any loss distribution, regardless of the domain. This reference explains what they measure, how they differ, why both matter for tail-driven decisions, and what practitioners should know about their limitations.

**2**

RISK MEASURES

**4**

COHERENCE AXIOMS

**P95 · P99**

TYPICAL LEVELS

**> 2× VaR**

CVAR HEAVY-TAIL SIGNAL

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## 01 Definitions

**VaR (Value-at-Risk)** reports the loss at a given percentile — the worst loss in the best  $(1-\alpha)$  fraction of scenarios. VaR at the 95th percentile ( $\text{VaR}_{95}$ ) answers: “What is the loss level that we exceed only 5% of the time?”

**CVaR (Conditional Value-at-Risk)**, also known as TVaR (Tail Value-at-Risk) or Expected Shortfall (ES), reports the average loss in the tail beyond VaR — the expected loss *given* that you are in the worst  $\alpha$  fraction of scenarios. CVaR at the 95th percentile ( $\text{CVaR}_{95}$ ) answers: “When losses exceed  $\text{VaR}_{95}$ , how bad are they on average?”

The distinction matters most when loss distributions have heavy tails, as cyber loss distributions typically do. Two distributions can have identical  $\text{VaR}_{95}$  but very different  $\text{CVaR}_{95}$  — one with a thin tail beyond  $\text{VaR}_{95}$ , one with extreme losses extending far beyond it. VaR cannot distinguish between them; CVaR can.

## 02 Why CVaR Matters: Coherence and Tail Sensitivity

Artzner, Delbaen, Eber, and Heath (1999) established four axioms that a risk measure should satisfy to be *coherent*:

- **Monotonicity.** If portfolio A always produces losses at least as large as portfolio B, then the risk measure for A should be at least as large as for B.
- **Translation invariance.** Adding a risk-free amount reduces the risk measure by exactly that amount.

- **Positive homogeneity.** Doubling the position doubles the risk.
- **Subadditivity.** The risk of a combined portfolio is no greater than the sum of individual risks — diversification should not increase measured risk.

VaR satisfies the first three but fails subadditivity for non-elliptical distributions — which includes the heavy-tailed distributions typical of cyber losses. This means VaR can produce the counterintuitive result that combining two portfolios appears riskier than holding them separately.

CVaR satisfies all four axioms (Rockafellar and Uryasev 2000, 2002). For organizations that aggregate risk across business units, scenarios, or model components, this mathematical property prevents internally contradictory risk rankings.

Beyond coherence, CVaR captures what VaR misses — the severity of tail losses. A loss distribution with  $\text{VaR}_{95} = \$10\text{M}$  might have  $\text{CVaR}_{95} = \$15\text{M}$  (thin tail) or  $\text{CVaR}_{95} = \$45\text{M}$  (heavy tail). For insurance limit decisions, capital adequacy, and risk appetite, the difference between these two scenarios is the difference between adequate and dangerously inadequate coverage.

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### 03 The Robustness Tradeoff

These theoretical advantages come with a practical cost. Cont, Deguest, and Scandolo (2010) demonstrate that CVaR has worse *qualitative robustness* than VaR: small perturbations in the tail of the fitted distribution produce larger changes in CVaR than in VaR.

For cyber risk, where tail data is sparse, this means CVaR amplifies the estimation instability that model risk validation is designed to detect. A CVaR estimate built on 12 extreme observations is uncertain and more fragile than the corresponding VaR estimate, because it integrates over precisely the region where the fitted distribution is least constrained by data.

This is not a reason to avoid CVaR. It is a reason to never report CVaR without sensitivity bands showing how the estimate changes under plausible alternative tail specifications, and to treat an unstable CVaR estimate as a signal that the tail is too uncertain to support the decision without compensating controls (conservative overlays, limits on use).

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### 04 Regulatory Direction

The regulatory trajectory favors tail-sensitive, coherent risk measures:

- **Basel III FRTB** replaced VaR with Expected Shortfall for market risk capital requirements, explicitly because VaR fails to capture tail risk.
- **Swiss Solvency Test (SST)** mandates TVaR rather than VaR.
- **PRA SS1/23** (UK, 2023) requires model risk management for all models used for business decision-making.

No cyber-specific regulation yet mandates CVaR/TVaR. But the direction is clear: regulators have moved to tail-sensitive measures wherever fat-tailed risk domains are subject to prudential oversight.

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## 05 Practical Recommendation: Report Both

VaR and CVaR/TVaR should be reported jointly as complementary diagnostics, not as preferred versus inferior measures:

- **VaR** provides a more robust (less estimation-sensitive) threshold estimate.
- **CVaR** provides a more informative (tail-shape-aware) conditional expectation.

The gap between them is the primary diagnostic:

- **Large VaR-CVaR gap** (e.g.,  $\text{CVaR} > 2 \times \text{VaR}$ ) signals heavy tail risk that VaR alone would conceal. Decision-makers need to understand that the losses in the tail are not just slightly worse than VaR — they are substantially worse.
- **Unstable CVaR** across plausible alternative specifications signals that the tail is too uncertain to support the decision without compensating controls.
- **Narrow gap** suggests a thinner tail where VaR and CVaR tell a similar story.

When a loss distribution model is used for insurance limit decisions, capital adequacy, or risk appetite, the validation (see *ERQI — Model Risk Testing & Analysis for Cyber Risk Quantification*) should assess:

- Whether the model produces both VaR and CVaR at relevant confidence levels (typically P95 and P99)
- Whether the VaR-CVaR gap is communicated to decision-makers
- Whether the CVaR estimate is stable enough to be actionable or whether it should trigger conservative overlays

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## 06 What These Measures Do Not Tell You

Both VaR and CVaR are estimates derived from a fitted model. They bound *recognized* uncertainty — the range of outcomes under the distributional assumptions the analyst selected. They do not bound:

- Unknown unknowns: model structures no one tested, failure modes no one imagined
- Systemic events outside the model's design basis
- Correlation structures the model does not capture

A model that reports  $\text{VaR}_{99} = \$50\text{M}$  and  $\text{CVaR}_{99} = \$120\text{M}$  is saying: “Under our assumptions, 1% of the time losses exceed \$50M, and when they do, they average \$120M.” It is not saying that \$120M is the worst case — the actual worst case depends on tail behavior that the model may not capture.

Decision-makers should understand that these measures represent “how wrong we think this could be given what we know,” not “the worst it could be.”

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